QUBO Formulation: Graph K-Coloring Problem

Graph K-Coloring Problem

Problem Statement. Given an undirected graph $G = (V, E)$ with $N = |V|$ vertices and a set of K available colors, assign exactly one color to each vertex such that no two adjacent vertices share the same color. The goal is to find a valid K-coloring if one exists, or otherwise minimize the number of monochromatic (violated) edges. A valid coloring corresponds to an objective value of zero.

Decision Variables. For each vertex $v \in V$ and each color $c \in \{0, 1, \ldots, K-1\}$, introduce a binary decision variable $x_{v,c} \in \{0, 1\}$. The variable $x_{v,c}$ equals 1 if and only if vertex v is assigned color c, and 0 otherwise.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) formulation by relaxing the hard constraints into the objective function via quadratic penalty terms.

Step 1 – Objective Function. The original goal is to minimize the number of violated edges. An edge $(u, v) \in E$ is violated if both endpoints receive the same color c. This is captured by the product $x_{u,c}x_{v,c}$. The objective to be minimized is:

$$\min \quad \sum_{(u,v)\in E} \sum_{c=0}^{K-1} x_{u,c}x_{v,c}. \tag{1}$$

Step 2 – Constraints. Each vertex must be assigned exactly one color. For every vertex $v \in V$, this hard constraint is expressed as:

$$\sum_{c=0}^{K-1} x_{v,c} = 1, \quad \forall v \in V. \tag{2}$$

Step 3 – Penalty Term Expansion. We enforce the constraints by adding a quadratic penalty to the objective. Let $\lambda > 0$ be the penalty weight. The penalty contribution for a single vertex v is:

$$P_v(\mathbf{x}) = \lambda \left(\sum_{c=0}^{K-1} x_{v,c} - 1 \right)^2. \tag{3}$$

Expanding the square yields:

$$\left(\sum_{c=0}^{K-1} x_{v,c} - 1\right)^2 = \sum_{c=0}^{K-1} x_{v,c}^2 + \sum_{c=0}^{K-1} \sum_{\substack{c'=0 \\ c' \neq c}}^{K-1} x_{v,c} x_{v,c'} - 2\sum_{c=0}^{K-1} x_{v,c} + 1. \tag{4}$$

Using the idempotent property of binary variables, $x_{v,c}^2 = x_{v,c}$, we simplify:

$$\left(\sum_{c=0}^{K-1} x_{v,c} - 1\right)^2 = \sum_{c=0}^{K-1} x_{v,c} + \sum_{c \neq c'} x_{v,c} x_{v,c'} - 2\sum_{c=0}^{K-1} x_{v,c} + 1 = \sum_{c \neq c'} x_{v,c} x_{v,c'} - \sum_{c=0}^{K-1} x_{v,c} + 1. \tag{5}$$

Summing over all vertices, the total penalty contribution to the Hamiltonian is:

$$P(\mathbf{x}) = \lambda \sum_{v \in V} \left(\sum_{c \neq c'} x_{v,c} x_{v,c'} - \sum_{c=0}^{K-1} x_{v,c} + 1 \right). \tag{6}$$

Step 4 – Combined Hamiltonian. Adding the objective and penalty terms gives the full QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = \sum_{(u,v) \in E} \sum_{c=0}^{K-1} x_{u,c} x_{v,c} + \lambda \sum_{v \in V} \left(\sum_{c \neq c'} x_{v,c} x_{v,c'} - \sum_{c=0}^{K-1} x_{v,c} + 1 \right). \tag{7}$$

Step 5 – Q Matrix Entries and Offset. We express $H(\mathbf{x})$ in the standard QUBO form $H(\mathbf{x}) = \sum_i Q_{ii} x_i + \sum_{i<j} 2Q_{ij} x_i x_j + c$, where i indexes the pair (v, c). Reading off the coefficients from the expanded Hamiltonian yields the following closed-form expressions for the Q matrix entries and constant offset:

$$Q_{(v,c),(v,c)} = -\lambda, \quad \forall v \in V,\, c \in \{0, \ldots, K-1\}, \tag{8}$$

$$Q_{(v,c),(v,c')} = \lambda, \quad \forall v \in V,\, c \neq c', \tag{9}$$

$$Q_{(u,c),(v,c)} = \frac{1}{2}, \quad \forall (u,v) \in E,\, c \in \{0, \ldots, K-1\}, \tag{10}$$

$$c = N\lambda. \tag{11}$$

All other entries Q_{ij} are zero.

Penalty Weight Justification. Let $\Delta = \max_{v \in V} \deg(v)$ denote the maximum vertex degree in G. Each variable $x_{v,c}$ participates in at most $\deg(v)$ terms in the objective function. Consequently, flipping a single variable can reduce the objective value by at most Δ. To guarantee that any constraint violation is strictly more costly than any potential objective improvement, we require $\lambda > \Delta$. Setting λ to any value strictly greater than Δ (e.g., $\lambda = \Delta + 1$) ensures that infeasible solutions are energetically unfavorable compared to all feasible solutions.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^{NK}$ recovers the optimal coloring because (i) any feasible assignment satisfies $\sum_c x_{v,c} = 1$ for all v, causing all penalty terms to vanish and reducing $H(\mathbf{x})$ exactly to the objective function counting violated edges; (ii) any infeasible assignment incurs a penalty of at least λ for each violated constraint. Since $\lambda > \Delta$, the penalty strictly exceeds the maximum possible decrease in the objective function achievable by any single variable flip. Therefore, the global minimum of $H(\mathbf{x})$ must occur at a feasible point, and among feasible points, $H(\mathbf{x})$ is minimized precisely when the number of violated edges is minimized.

Illustrative Example. Consider a concrete instance with $N = 3$ vertices, $K = 3$ colors, and a complete graph $G = K_3$ where every pair of vertices is connected. The edges are $(0,1)$, $(1,2)$, and $(2,0)$. The maximum degree is $\Delta = 2$, so we set $\lambda = 10$. The variable mapping is $i = vK + c$, yielding indices $0,1,2$ for vertex 0, $3,4,5$ for vertex 1, and $6,7,8$ for vertex 2.

The concrete objective counts monochromatic edges:

$$\text{Obj}(\mathbf{x}) = \sum_{c=0}^{2} \left(x_{0,c}x_{1,c} + x_{1,c}x_{2,c} + x_{2,c}x_{0,c} \right). \tag{12}$$

The concrete penalty terms for each vertex $v \in \{0,1,2\}$ are:

$$P_v(\mathbf{x}) = 10 \left(\sum_{c \neq c'} x_{v,c}x_{v,c'} - \sum_{c=0}^{2} x_{v,c} + 1 \right). \tag{13}$$

The resulting Hamiltonian $H(\mathbf{x})$ is the sum of the objective and all three penalty terms. The non-zero Q matrix entries for this instance are:

$$Q_{0,0} = Q_{1,1} = Q_{2,2} = Q_{3,3} = Q_{4,4} = Q_{5,5} = Q_{6,6} = Q_{7,7} = Q_{8,8} = -10, \tag{14}$$

$$Q_{0,1} = Q_{0,2} = Q_{1,2} = Q_{3,4} = Q_{3,5} = Q_{4,5} = Q_{6,7} = Q_{6,8} = Q_{7,8} = 10, \tag{15}$$

$$Q_{0,3} = Q_{1,4} = Q_{2,5} = Q_{3,6} = Q_{4,7} = Q_{5,8} = Q_{6,0} = Q_{7,1} = Q_{8,2} = \frac{1}{2}, \tag{16}$$

with offset $c = 3 \times 10 = 30$.

A valid 3-coloring exists. Setting $x_{0,0} = 1$, $x_{1,1} = 1$, $x_{2,2} = 1$ (all other variables 0) satisfies the exactly-one-color constraints and yields zero violated edges. The optimal energy is $H(\mathbf{x}^*) = 0 + 30 = 30$. Any assignment violating the constraints incurs a penalty of at least 10, strictly exceeding the maximum possible objective reduction of 2, confirming the formulation's correctness for this instance.